



UNIVERSITY OF VERONA

DEPARTMENT OF COMPUTER SCIENCE · BACHELOR'S DEGREE
PROGRAM

BACHELOR'S THESIS

Learning, coordination and generalization in Multi-Agent Reinforcement Learning: a comparison between IPPPO, MAPPO and MASAC

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Coordination and generalization

CONTEXT

Multiple autonomous agents must coordinate in a shared environment, without communication or a central controller.

CHALLENGES

Non-stationarity agents' behavior changes at every update, the target to optimize keeps moving.

Coordination without communication, no central controller.

Generalization across multiple agents and different topologies.

Research question and objective

Approaches: decentralized (**IPPO**), centralized (**MAPPO**), off-policy (**MASAC**) — how do they differ in training performance and generalization ability?

Objective: training on a fixed configuration, zero-shot evaluation on increasing densities and on a never-before-seen topology.

IN THIS PRESENTATION

The limits of the MLP architecture and the impact of limiting observation size on generalization ability.

Similarities and differences in performance between a centralized and a decentralized approach.

The failure of an off-policy MASAC approach in a non-stationary, chaotic environment with discrete actions.

The general framework: Markov Game

MDP extended to multiple actors

$$\langle S, N, \{A_n\}, P, \{R_n\}, \gamma, \mu \rangle$$

S finite number of states

N finite number of agents

A_n actions of the single agent · joint action $A = A_1 \times \dots \times A_n$

P transition, depends on the joint action: $S \times A_1 \times \dots \times A_n \rightarrow \Delta(S)$

R_n reward for each agent: $S \times A_1 \times \dots \times A_n \rightarrow [-1, 1]$

γ discount factor

μ $\in \Delta(S)$, initial state distribution

NON-STATIONARITY

In MARL multiple agents learn simultaneously in the same environment. Each agent is part of the others' environment.

From the individual's perspective, the environment becomes **non-stationary**: the dynamics of the environment change at every update

Our case: Partially Observable Markov Game

Partial observability	each agent does not see the global state S , but only a local observation O_n . The Markov Game becomes a multi-agent POMDP.
Decentralized execution	each agent decides based only on its own observation, without a central controller.
Individual objective	Same type of objective (reaching the destination), but each agent optimizes its own return. Global welfare is a consequence, not the target.
Individual reward	$R_n : S \times A_1 \times \dots \times A_n \rightarrow [-1, 1]$ no shared reward.

$$\langle S, N, \{A_n\}, P, \{R_n\}, \{O_n\}, \gamma \rangle$$

FORMAL DESCRIPTION

General-sum game with partial observability and decentralized execution, in which cooperation must **emerge** , it is not imposed by the reward signal.

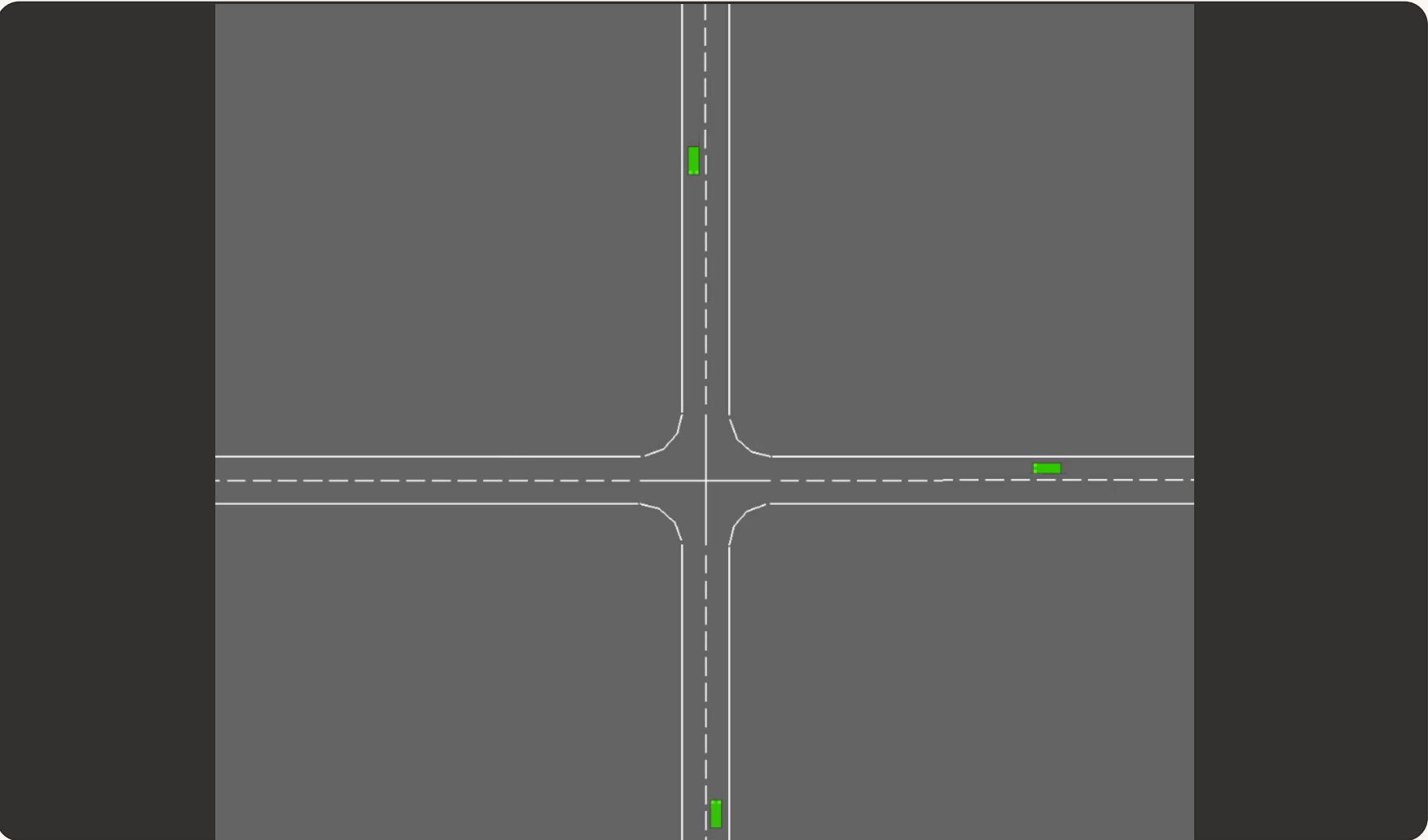
Highway-Env

decision-making for autonomous driving · minimalist

NB · UNSIGNALLED SCENARIOS
no explicit right-of-way logic

Intersection

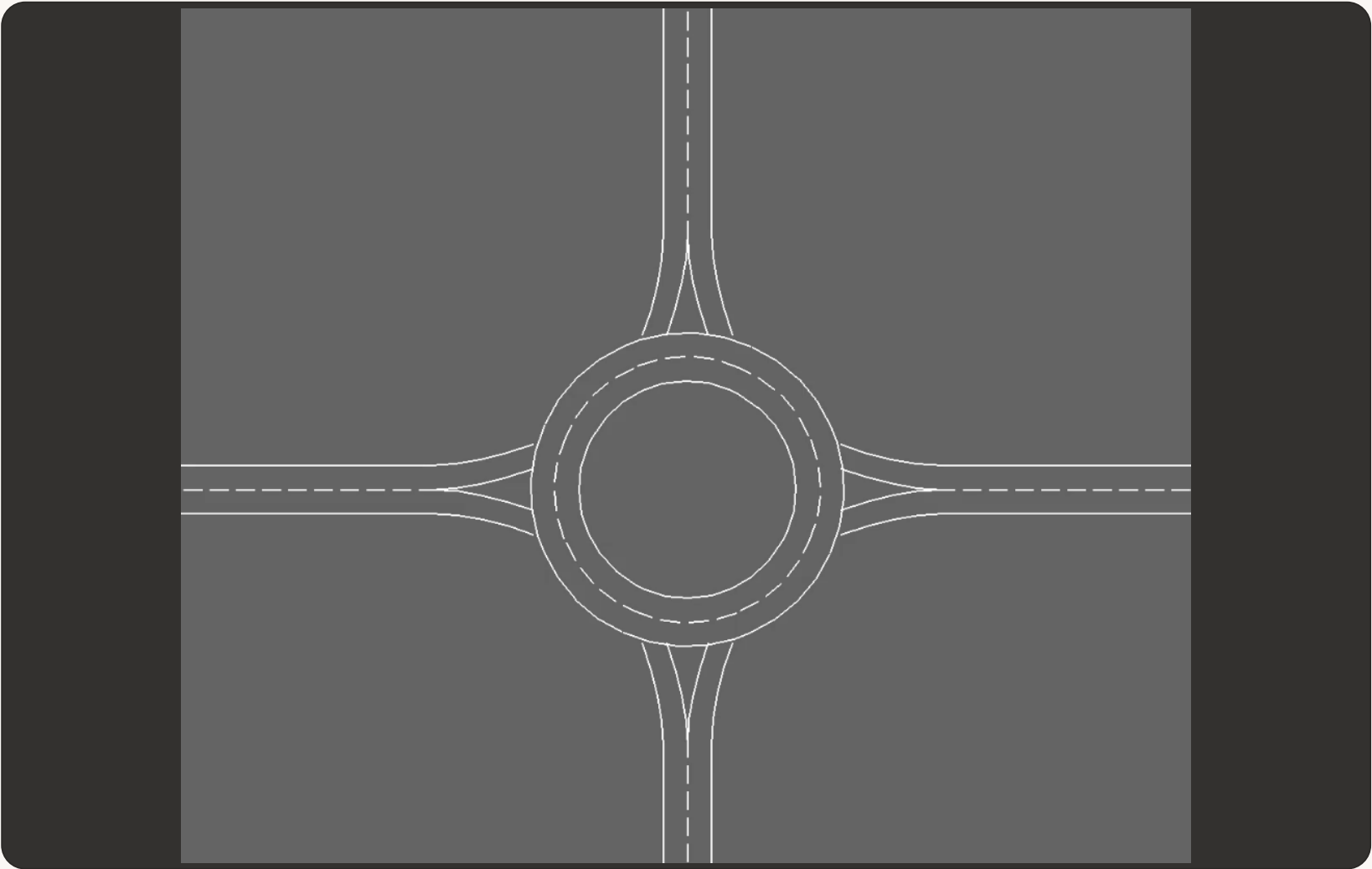
CHAOTIC · REACTIVE



Gap acceptance. Decision reactive and local. **Low interdependence.**

Roundabout

STRUCTURED · SOCIAL



Yielding right of way. Cooperative decision: immediate cost for collective benefit. **High interdependence.**

Training configuration

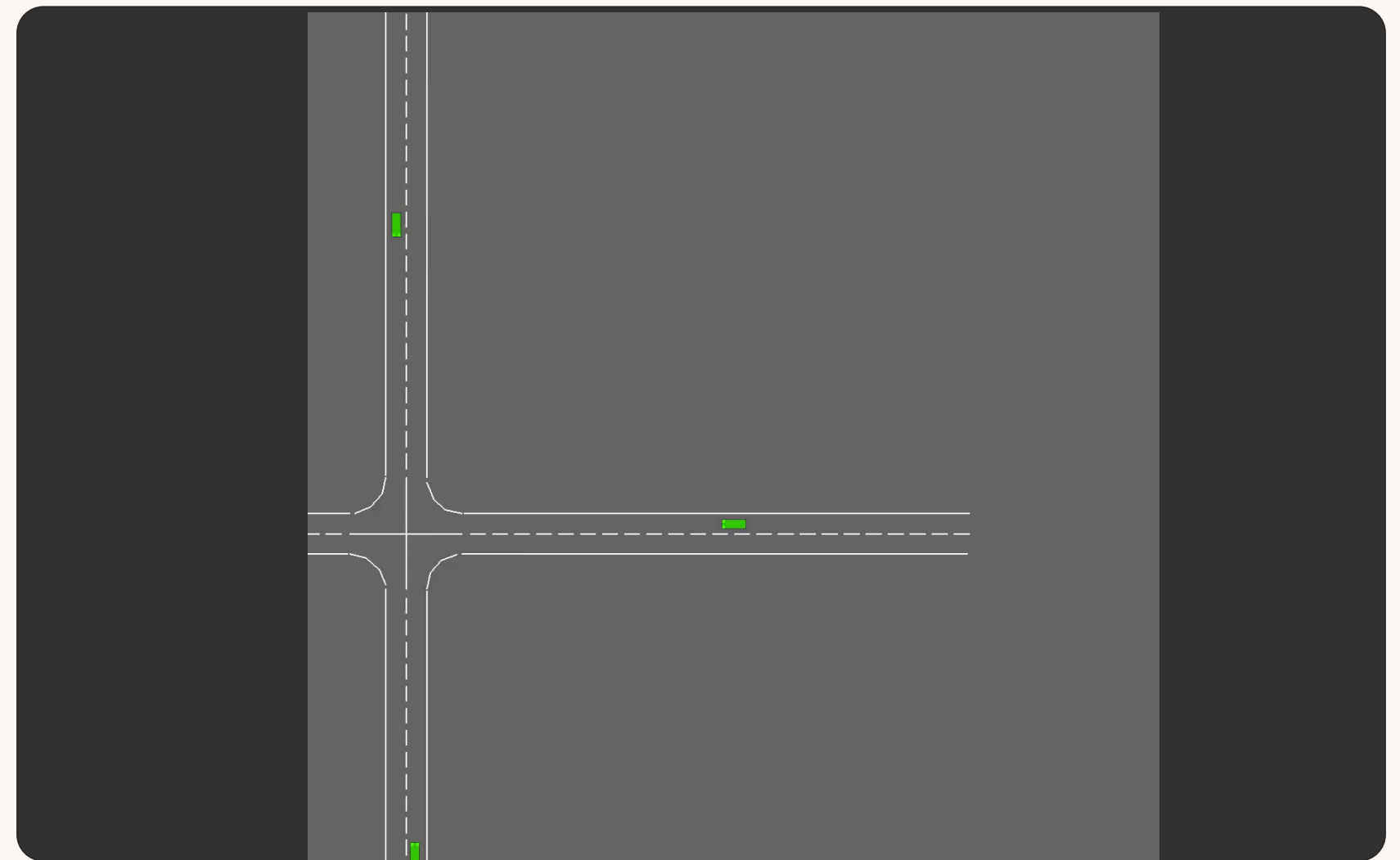
- training **only on the intersection**
- 3 agents
- random start and destination points at every reset
- roundabout used only in evaluation (*cross-domain test*)

ZERO-SHOT EVALUATION

As the **density of vehicles increases**

As the **topology** varies (roundabout introduced)

INTERSECTION · 3 AGENTS



Observations

Kinematics $V \times F$

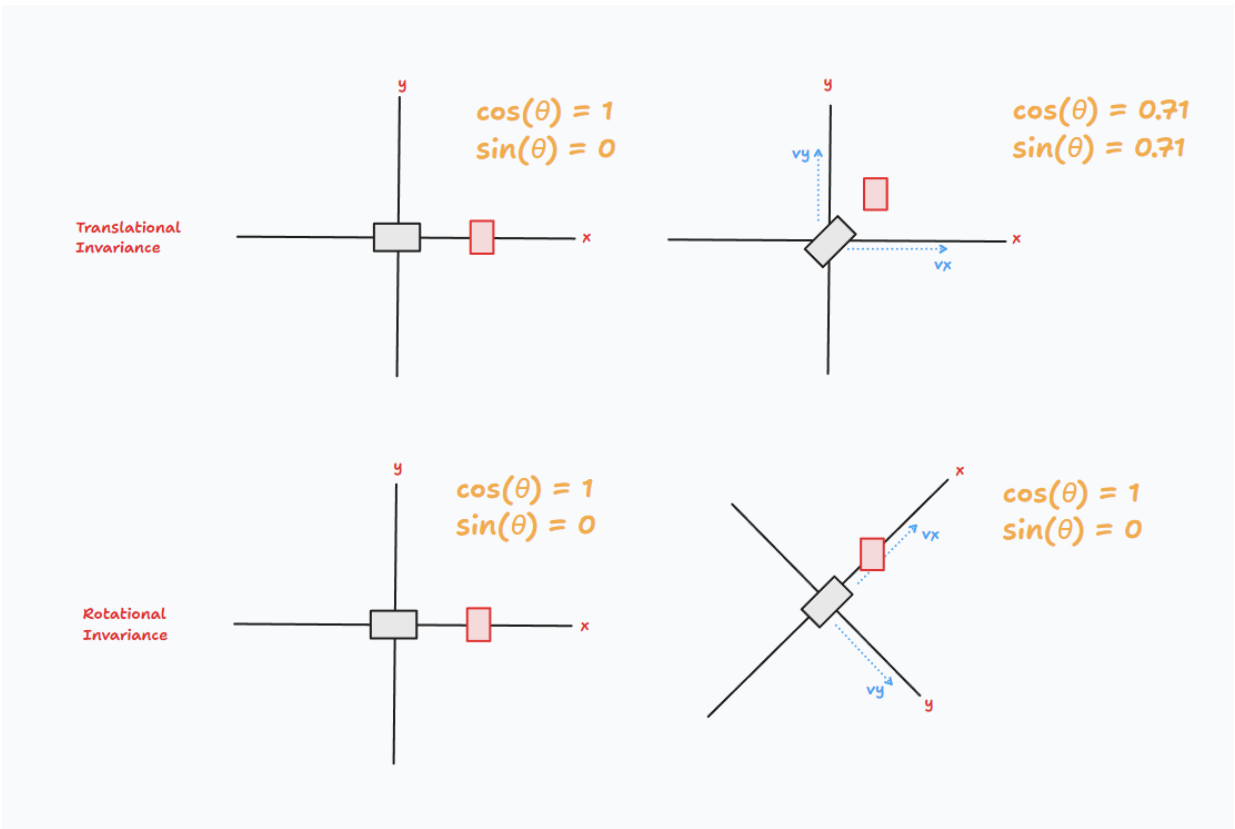
$V = 3$

ego + 2 nearest vehicles, 360° vision. Beyond 3 vehicles → partial observability.

$F = 8$

presence · x · y · vx · vy cos_h · sin_h (heading) cos_d · sin_d
(goal)

Completely relative to the ego → **translational and rotational** invariance.



In evaluation with density of 4+ agents, the size stays 3: each agent becomes **blind beyond the 2 nearest vehicles**.

Actions and reward

ACTIONS

Discrete, high-level · no steering control, only acceleration

- 0 SLOWER

slow
down
- 1 IDLE

maintain
- 2 FASTER

accelerat
e

Speed: [0, 1.5, 3, 4.5, 6] m/s

SPARSE REWARDS

Normalized to [-1, 1]

arrived_reward	+50
collision_reward	-50
step_penalty	-0.05

Individual reward, no team signal: **cooperation is not incentivized by a reward.**

Implementation of the Centralized Critic

- Framework: **Ray RLlib**, standard for distributed RL
- **IPPO** = default multi-agent implementation of PPO
- **MAPPO** = custom class `CentralizedCriticModel` (PyTorch)

FORWARD()

```
logits = self.actor(obs)
value = self.critic(global_state)
```

EXECUTION · DECENTRALIZED

local obs o_i



Actor $\pi(a_i | o_i)$



action a_i

Local observations only: used in training **and** execution

TRAINING · CENTRALIZED

$s = [o_1 \parallel o_2 \parallel o_3]$



Critic $V(s)$



advantage → actor update

Global state: used **only** in training

Shared baseline

IPPO and MAPPO share the same hyperparameters: the only difference is the critic.
Search via **Optuna**.

IPPO · MAPPO

```
gamma = 0.975
clip_param = 0.2
lr = 3e-4 entropy_coeff = 0.01
use_gae = True lambda = 0.95
```

MASAC: hyperparameters from literature standards. Manual tuning, less exhaustive.

MASAC

```
gamma = 0.975
train_batch_size_per_learner = 512
actor_lr = 1e-4 critic_lr = 3e-4
alpha_lr = 1e-4 initial_alpha = 0.1
num_steps_sampled_before_learning_starts = 20000
target_entropy = 0.8 tau = 0.005
buffer_capacity = 500000
```

The only difference between IPPO and MAPPO is the **critic**: same code, same hyperparameters.

Reproducibility and evaluation

REPRODUCIBILITY

- 3 seed different: statistical validity, variance measurement
- 200 episodes per density level, with identical seeds for IPPO, MAPPO and MASAC
- evaluation with **greedy actions**

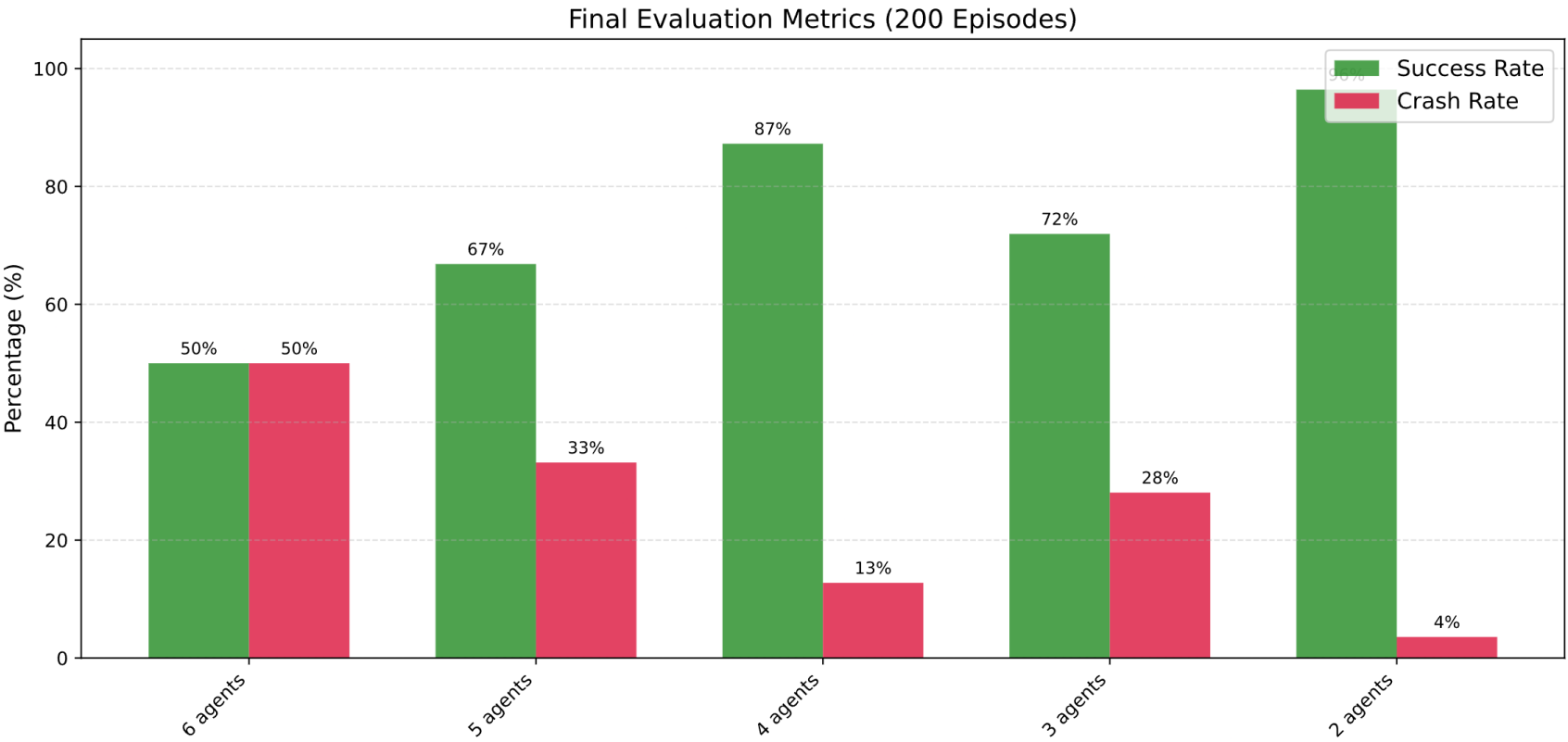
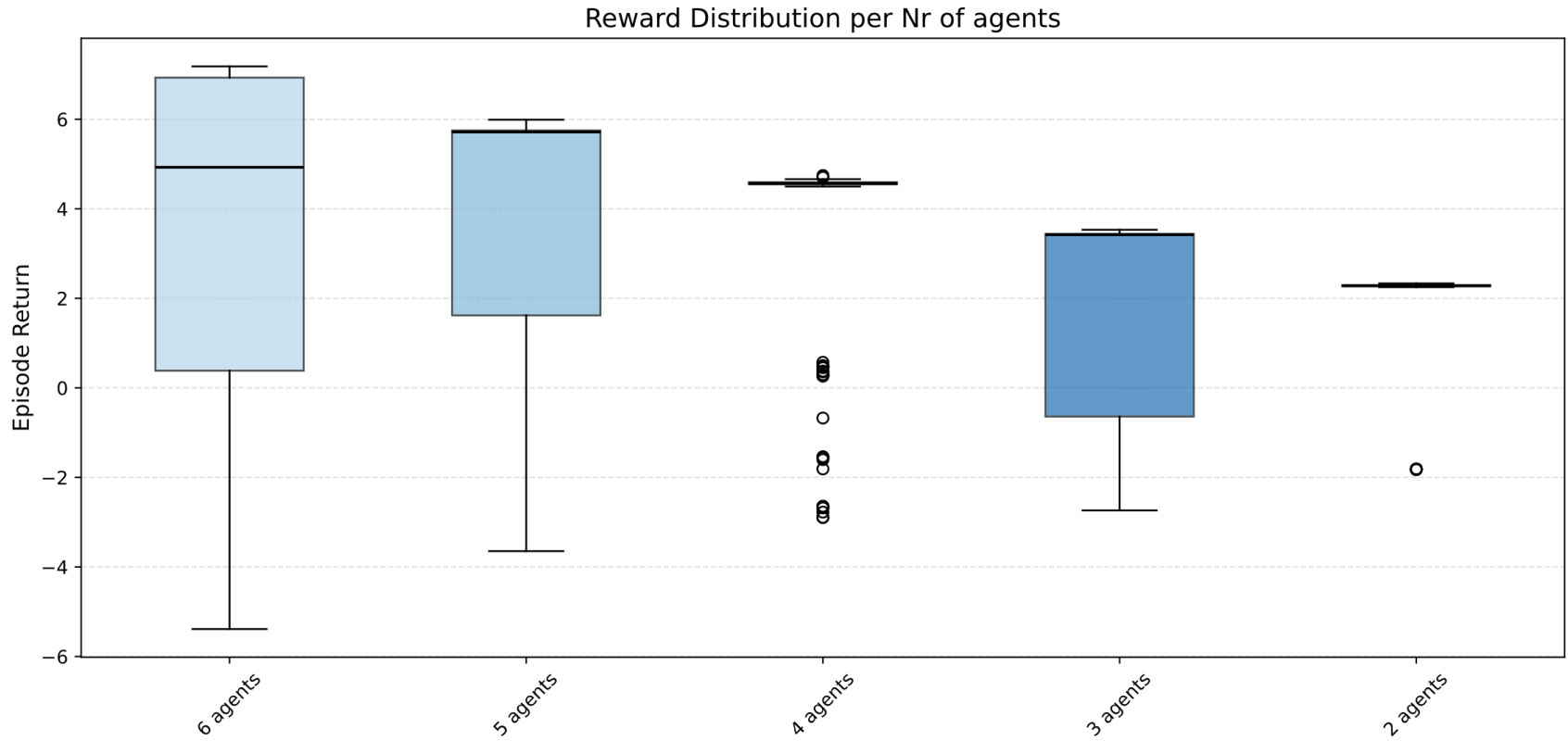
REFERENCE METRICS

- IPPO / MAPPO (on-policy) `vf_loss · episode_return_mean · entropy`
- MASAC (off-policy) `alpha_loss · critic_loss · actor_loss · episode_return_mean`
- Common success rate and crash rate per density level

Truncation vs zero-padding: the MLP's limit

vehicles_count = 4 truncated observation

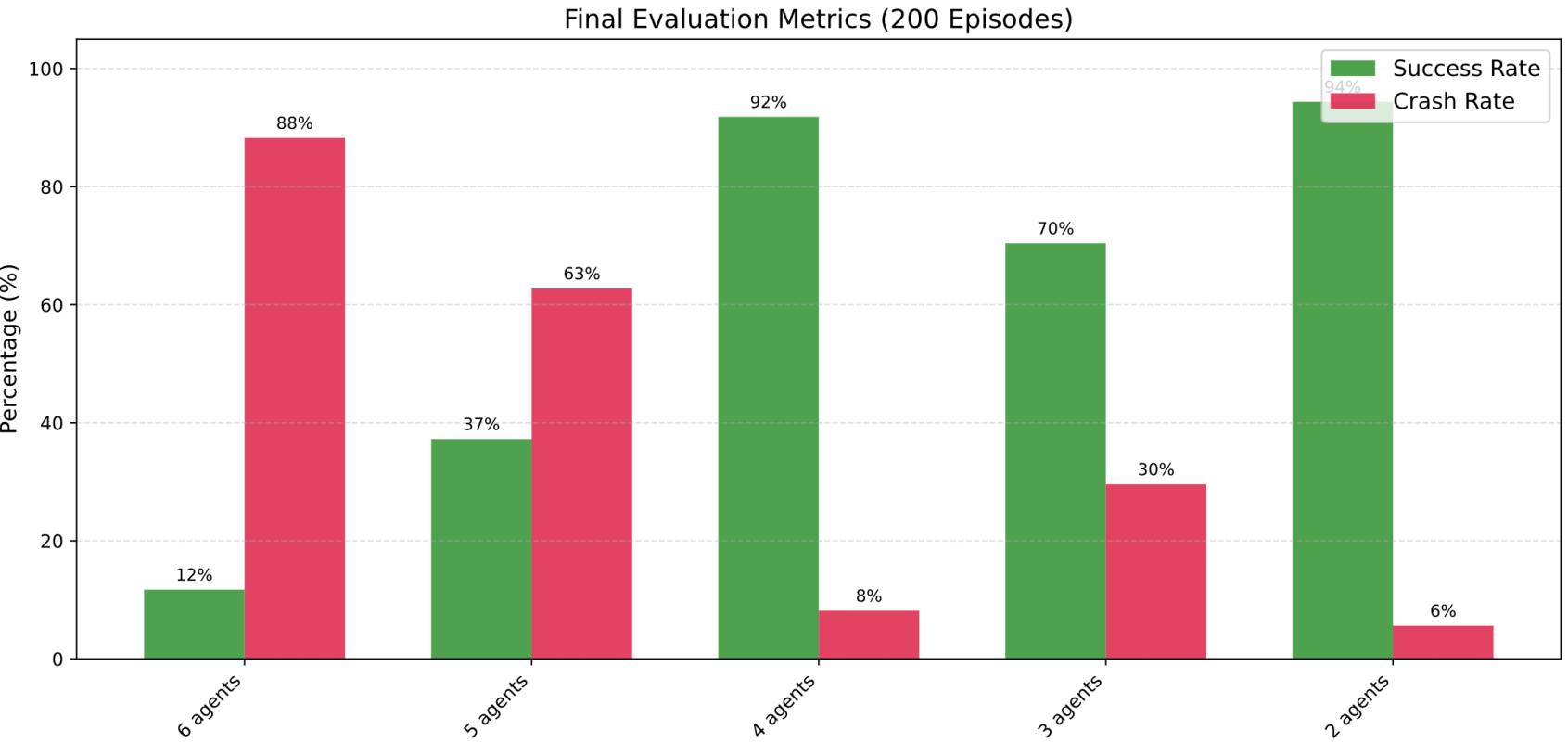
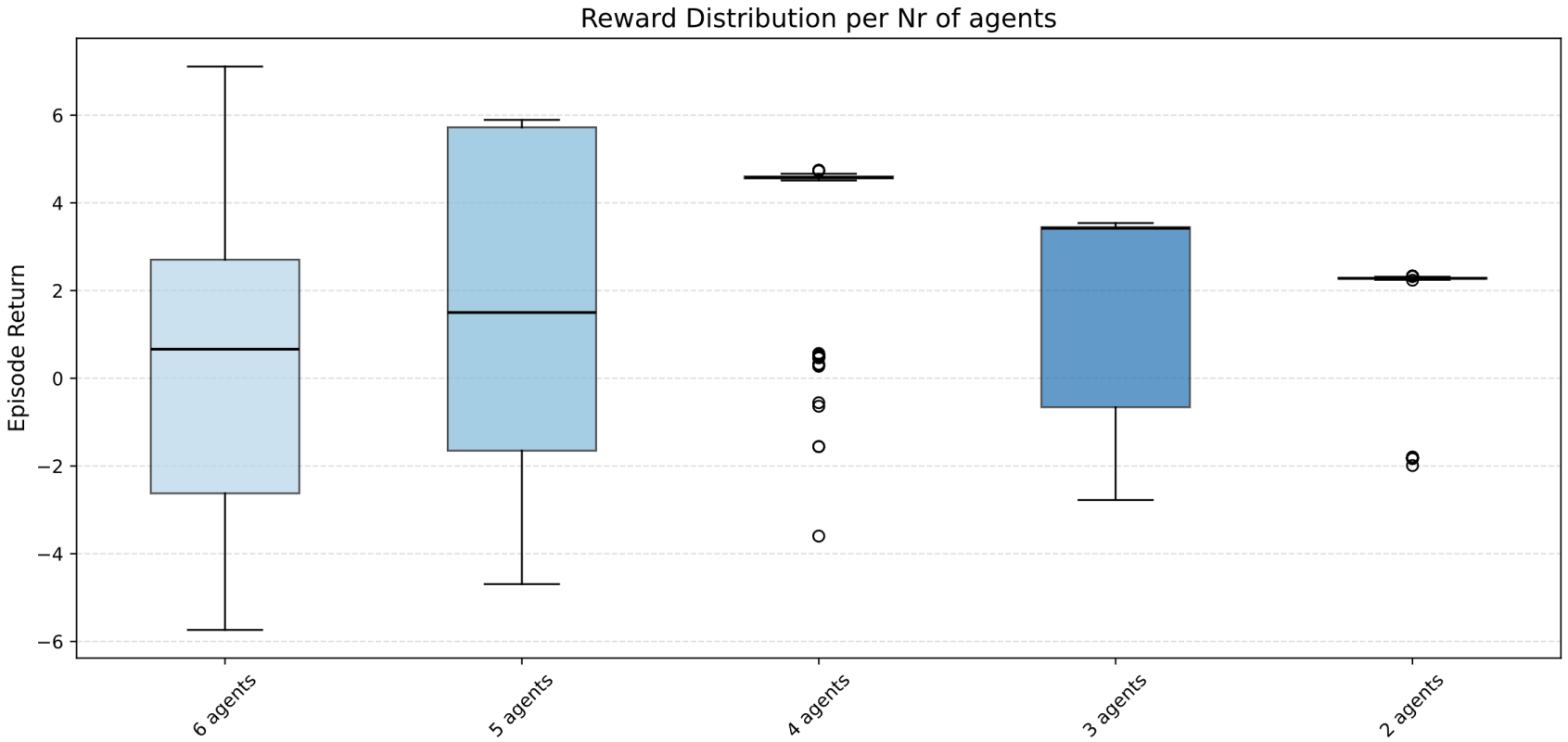
Zero-Shot Generalization Comparison



Rows always valid → **gradual and predictable degradation.**

vehicles_count = 6 slots trained to zero

Zero-Shot Generalization Comparison



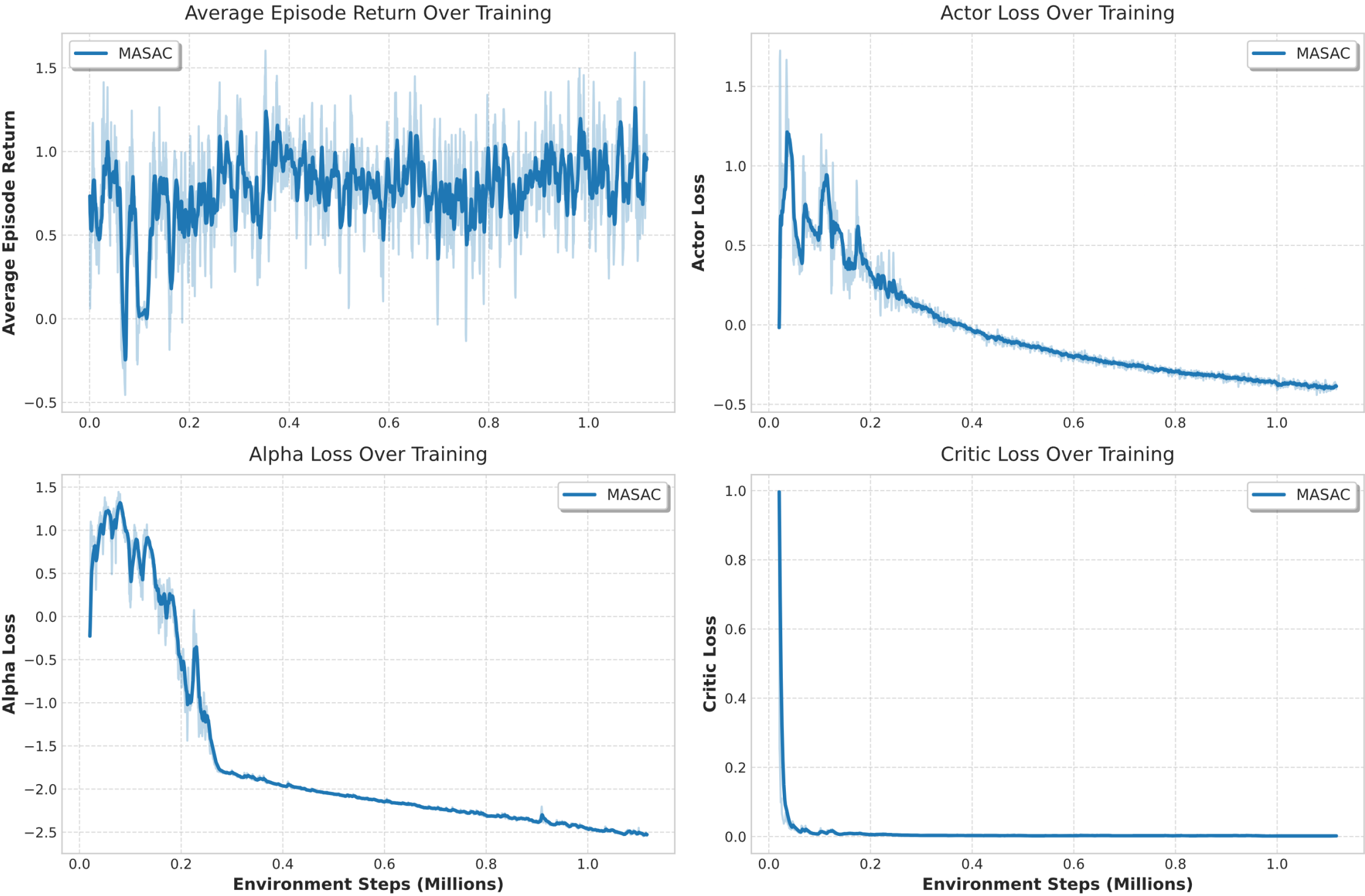
Zero slots filled by real vehicles → **sharp degradation.**

RESULTS

Results MASAC

The off-policy does not converge: instability

Multi-Agent Highway Intersection (MASAC Failure)



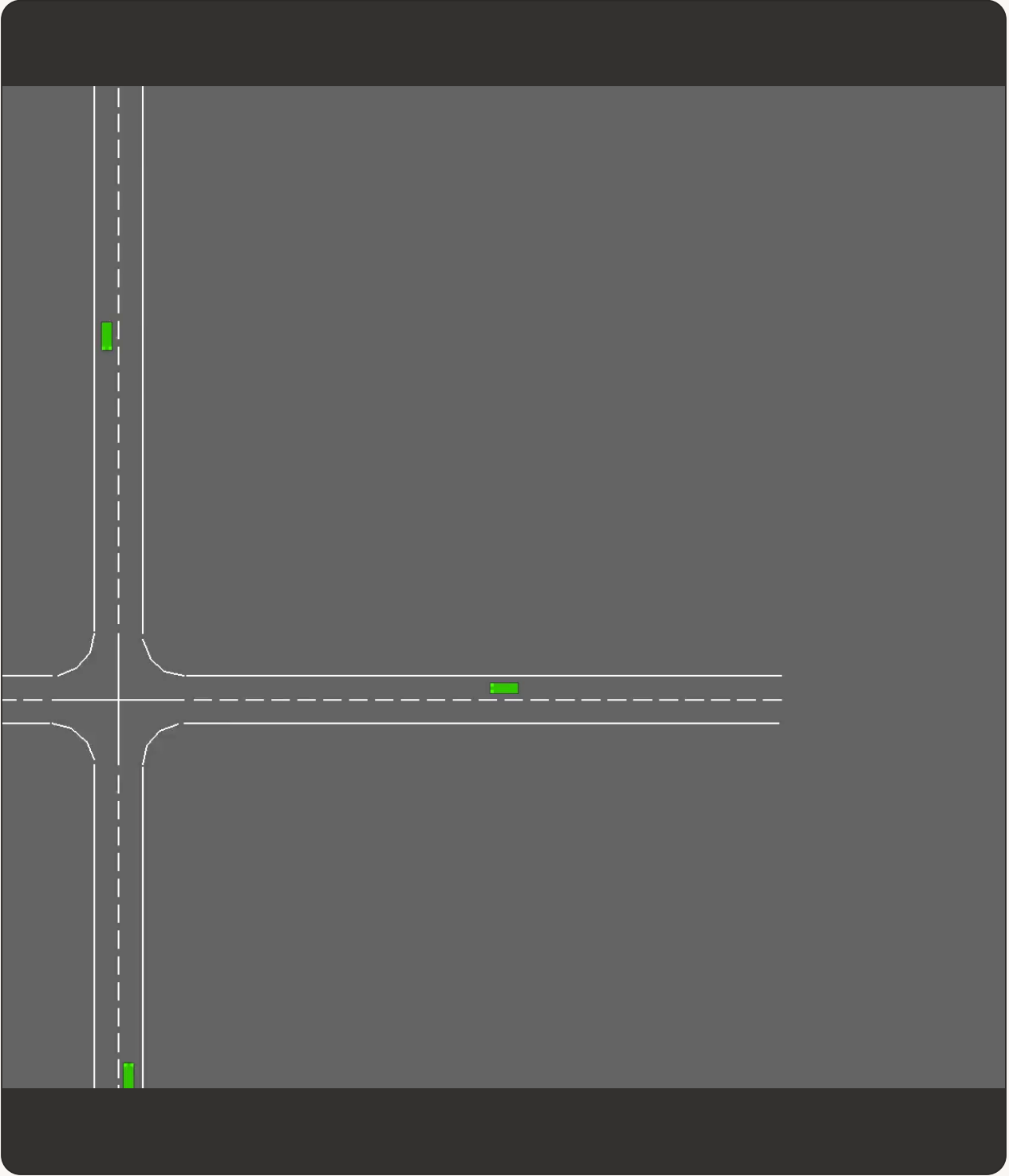
alpha_loss flattens out

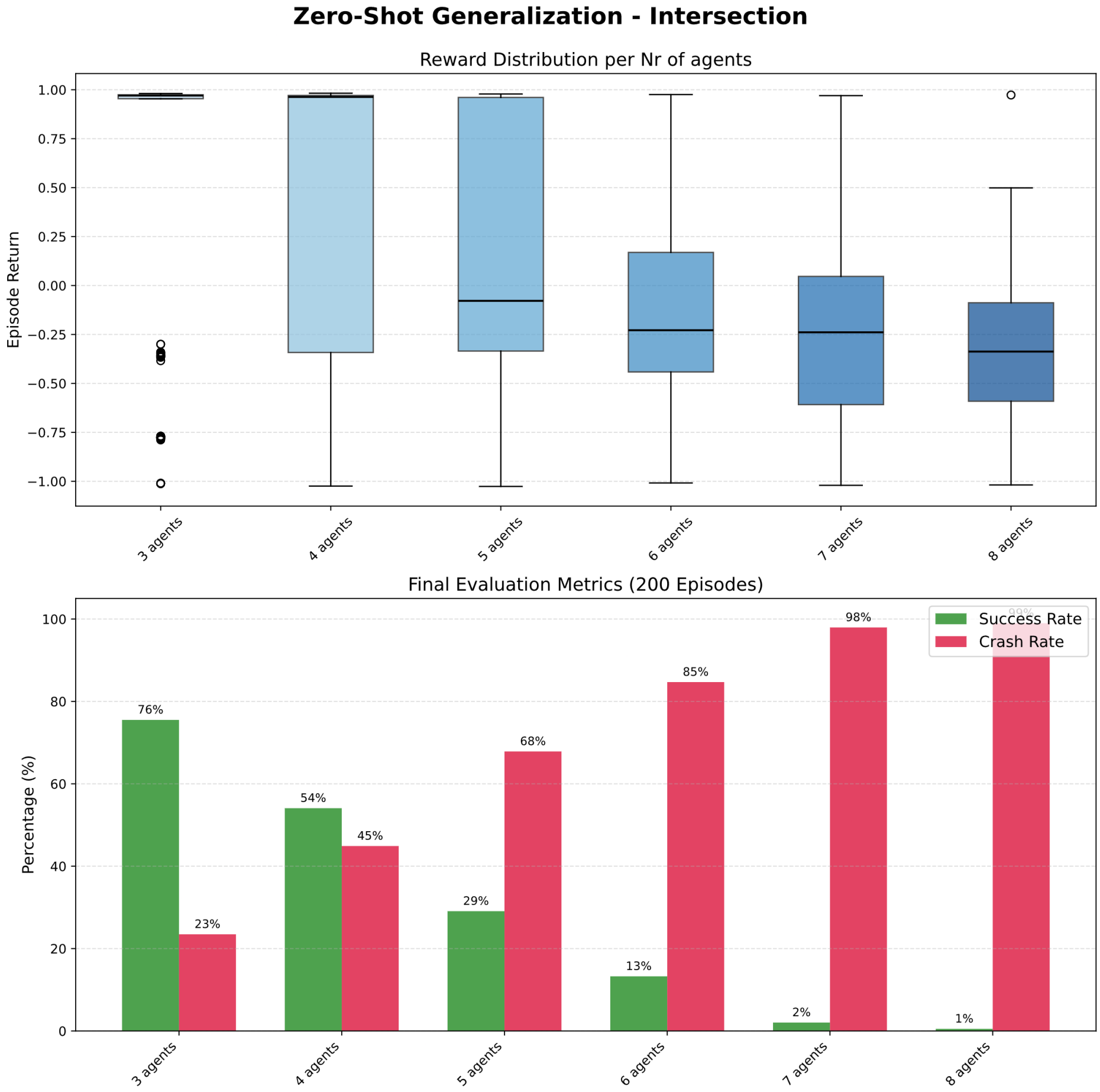
Exploration is not rewarded, fatal for MASAC; suboptimal policy

MAX SUCCESS RATE

~76%

Suboptimal policy



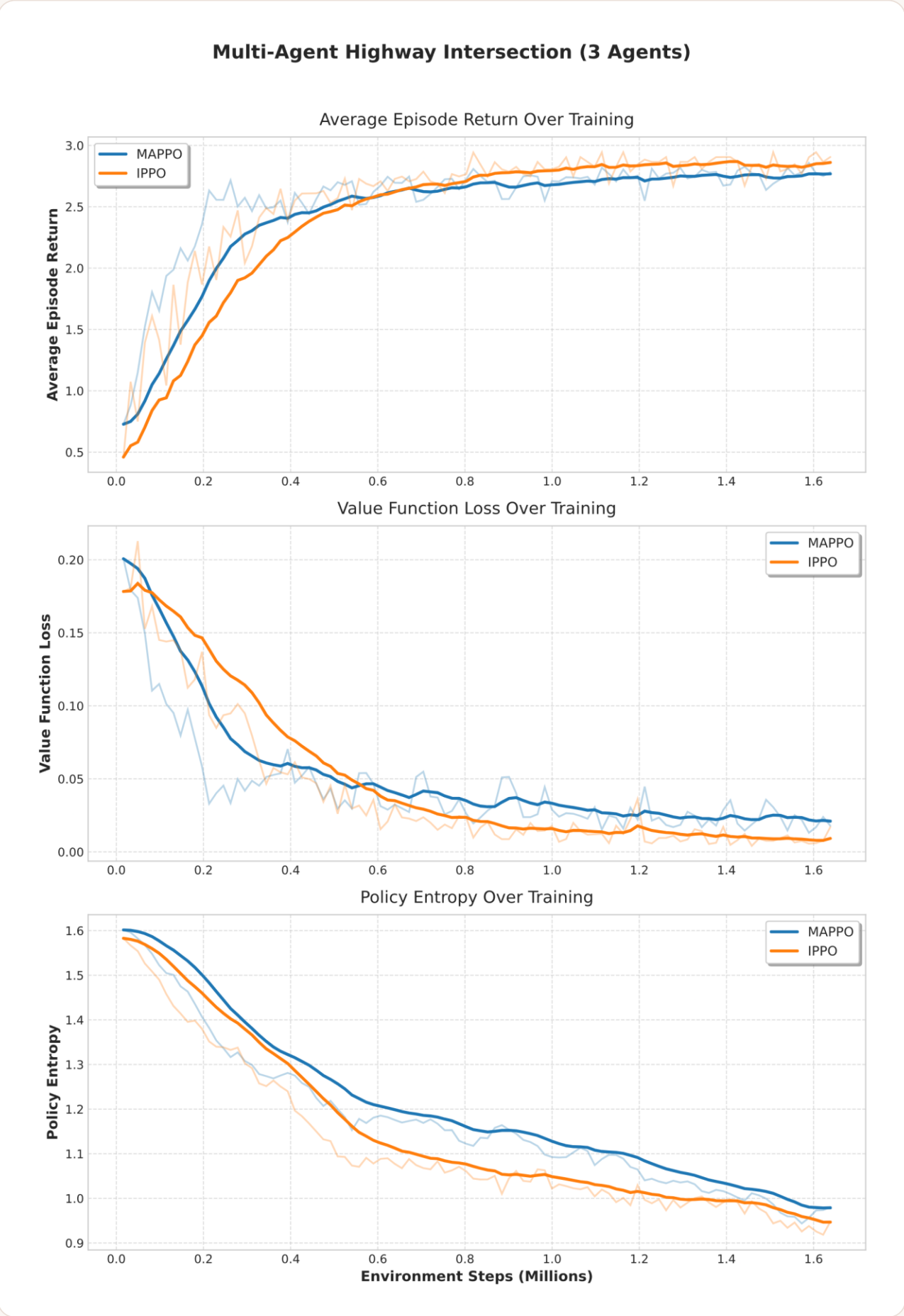


RESULTS

Results IPPO vs MAPPO

Both on-policy methods converge

MAPPO IPPO



FINAL SUCCESS RATE (3 SEEDS)

IPPO
≈ 0.97

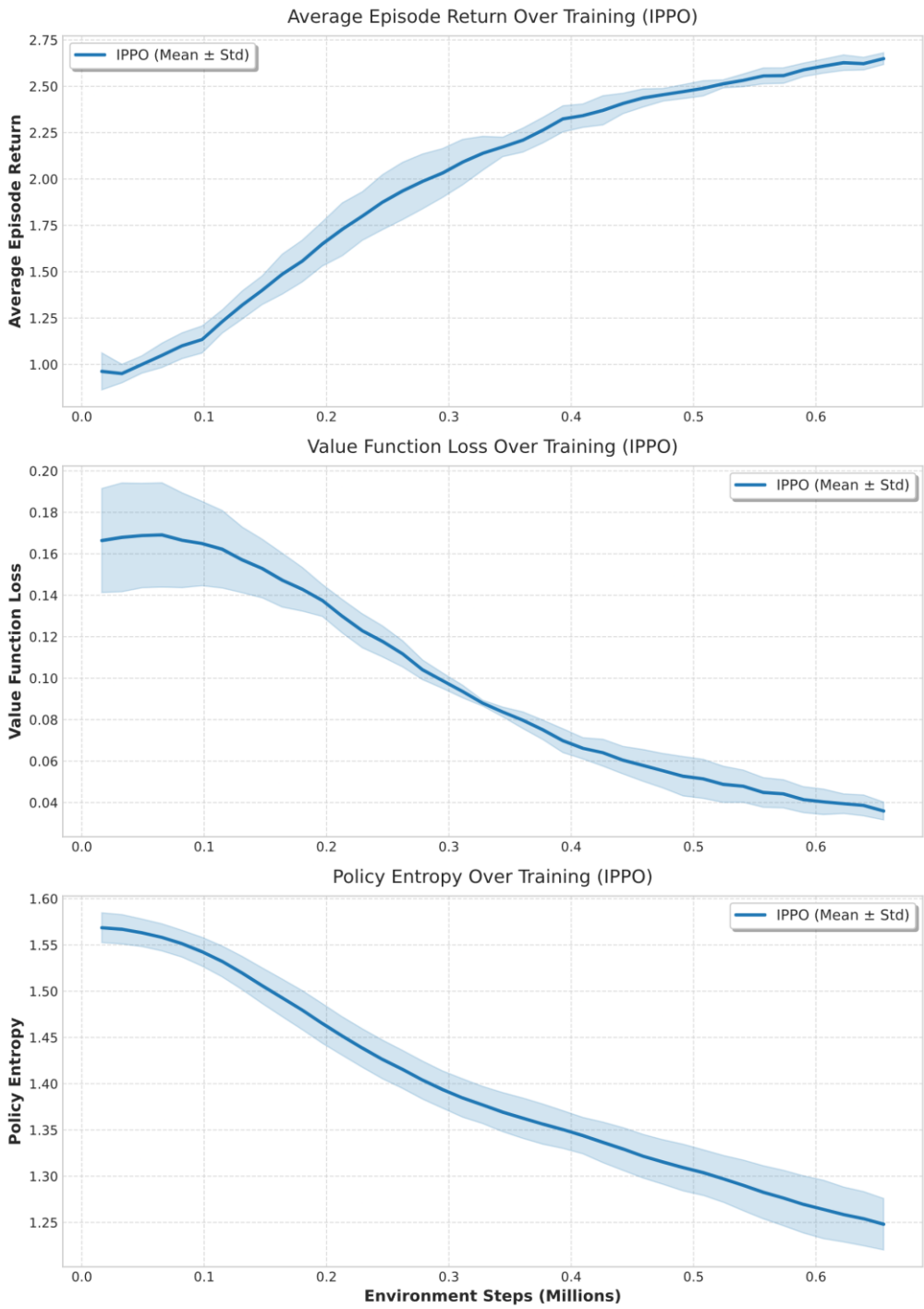
MAPPO
≈ 0.97

- Reliable convergence across all 3 seeds.
- Final performance is substantially equivalent on the training intersection.

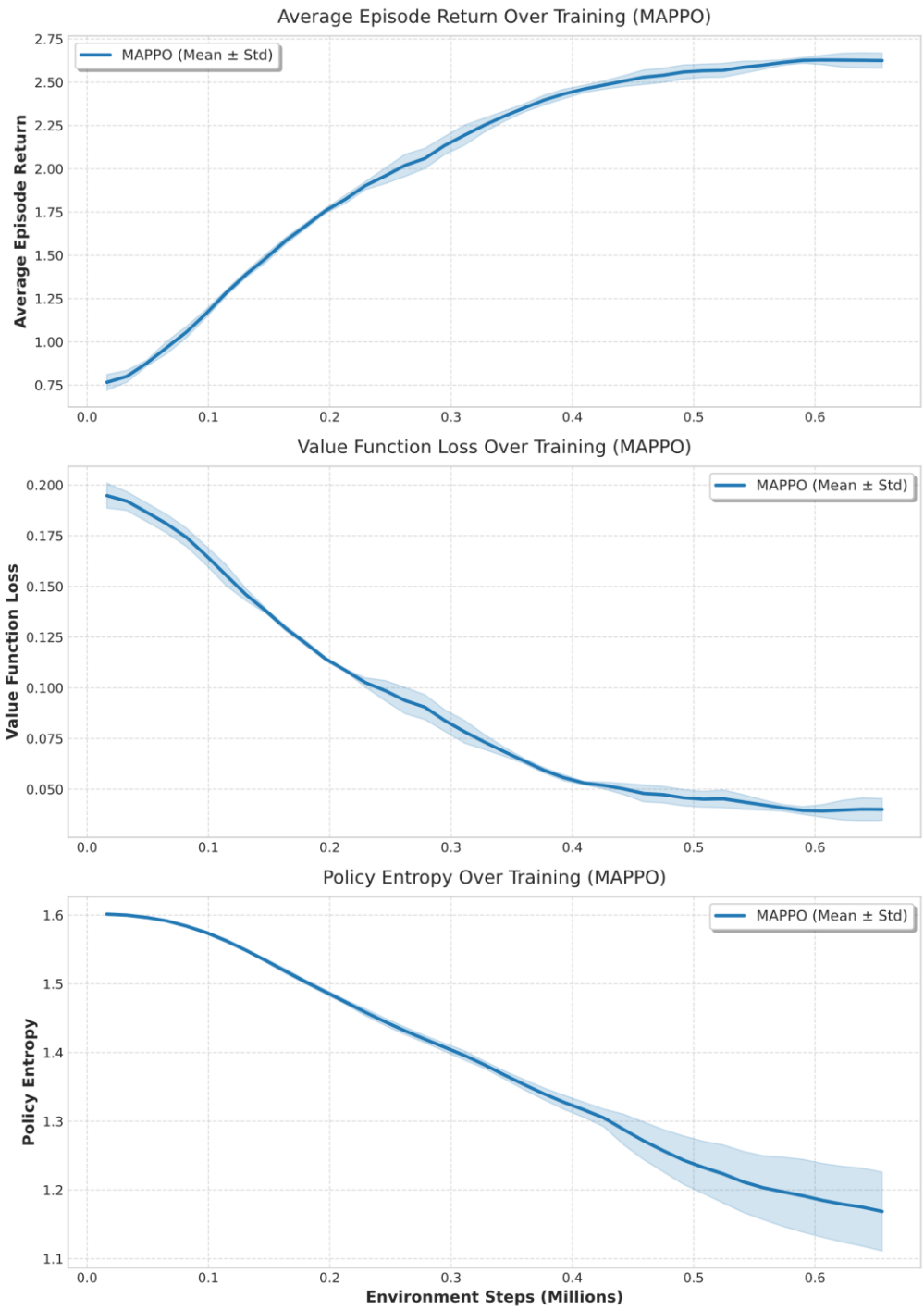
IPPO decentralized

MAPPO centralized

IPPO Performance Across Seeds



MAPPO Performance Across Seeds



INITIAL VF-LOSS STD

IPPO
≈ 0.023

MAPPO
≈ 0.010

~2.3×

IPPO is more sensitive to the seed: estimating value from local observations under non-stationarity is harder.

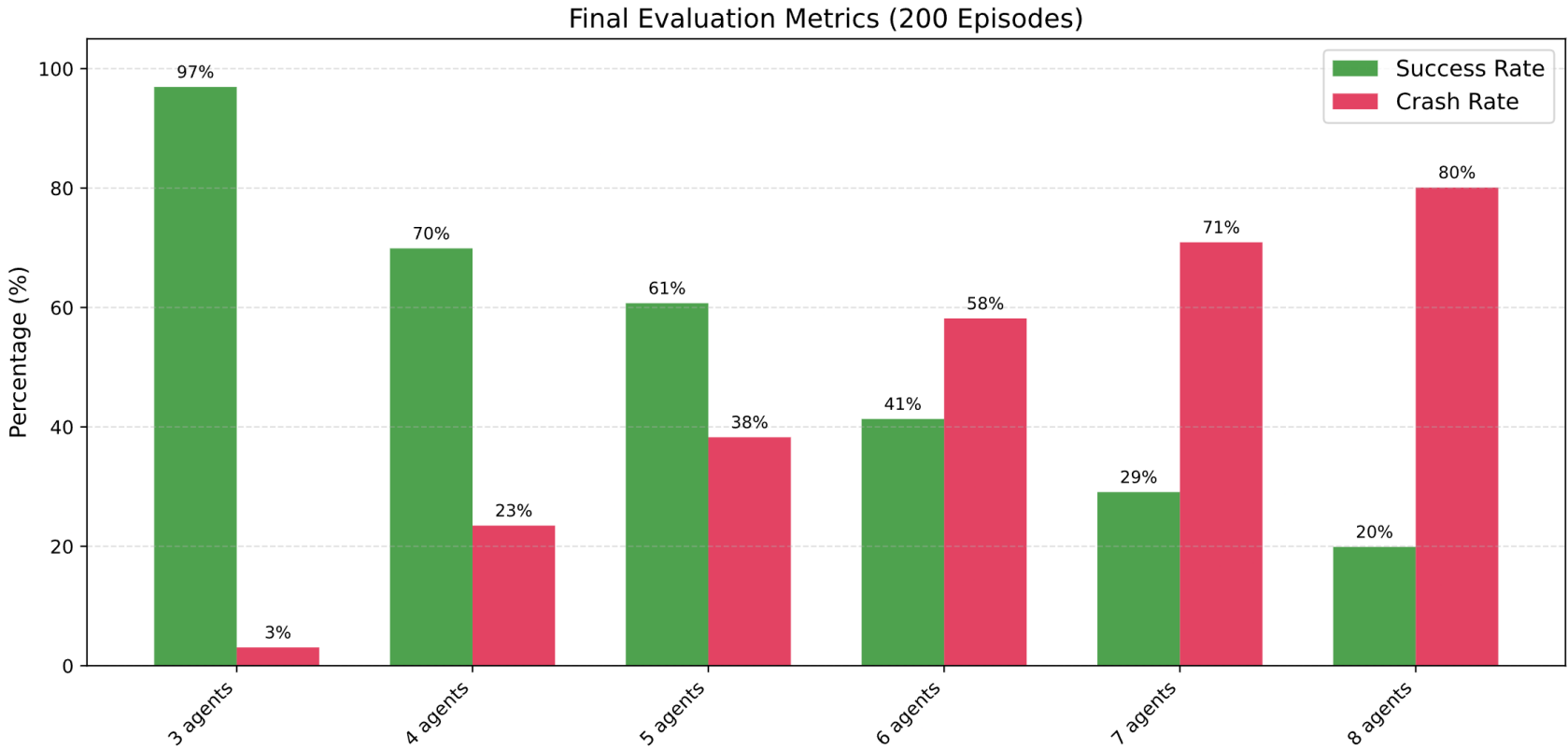
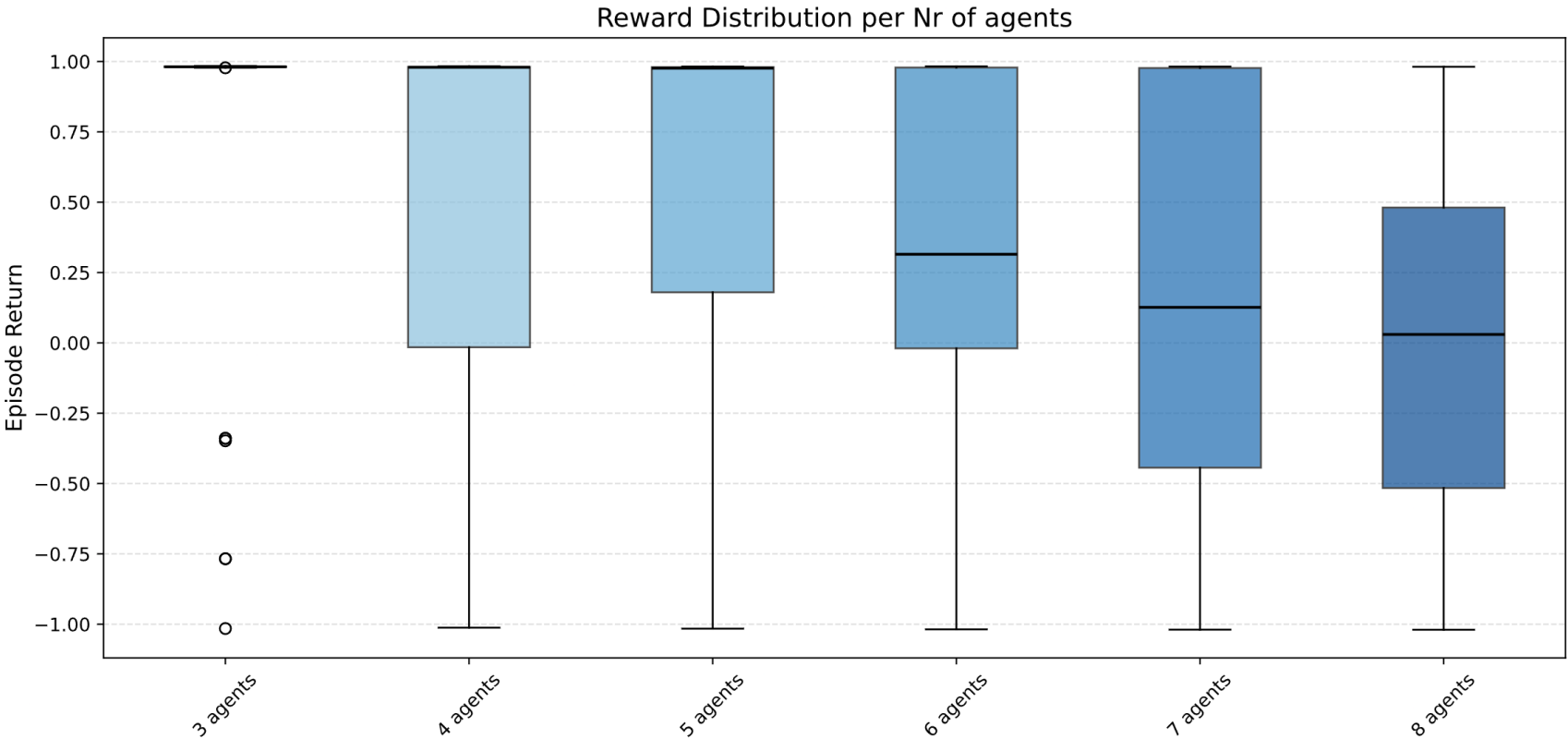
IPPO vs MAPPO

Evaluation on Intersection

On the intersection the two methods are equivalent

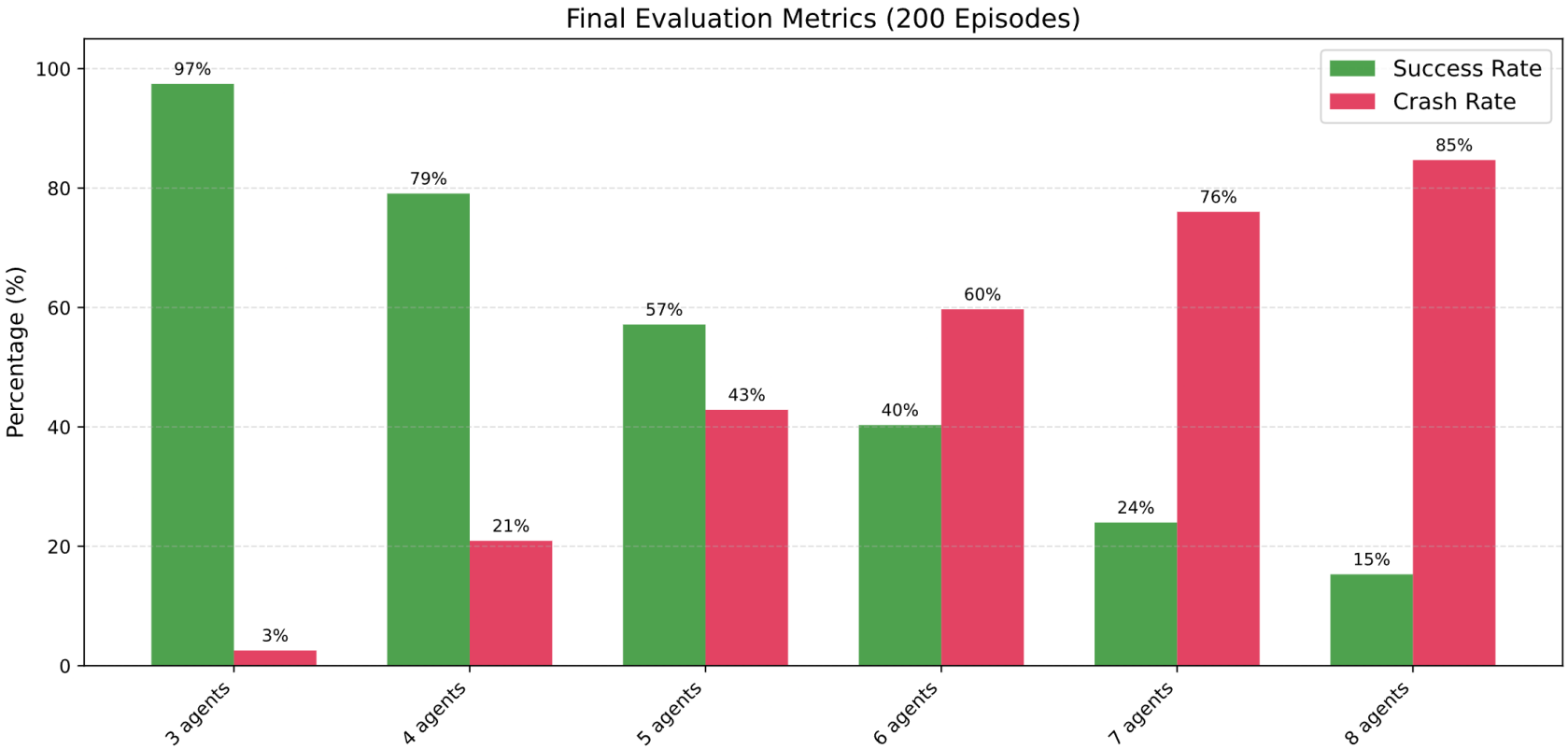
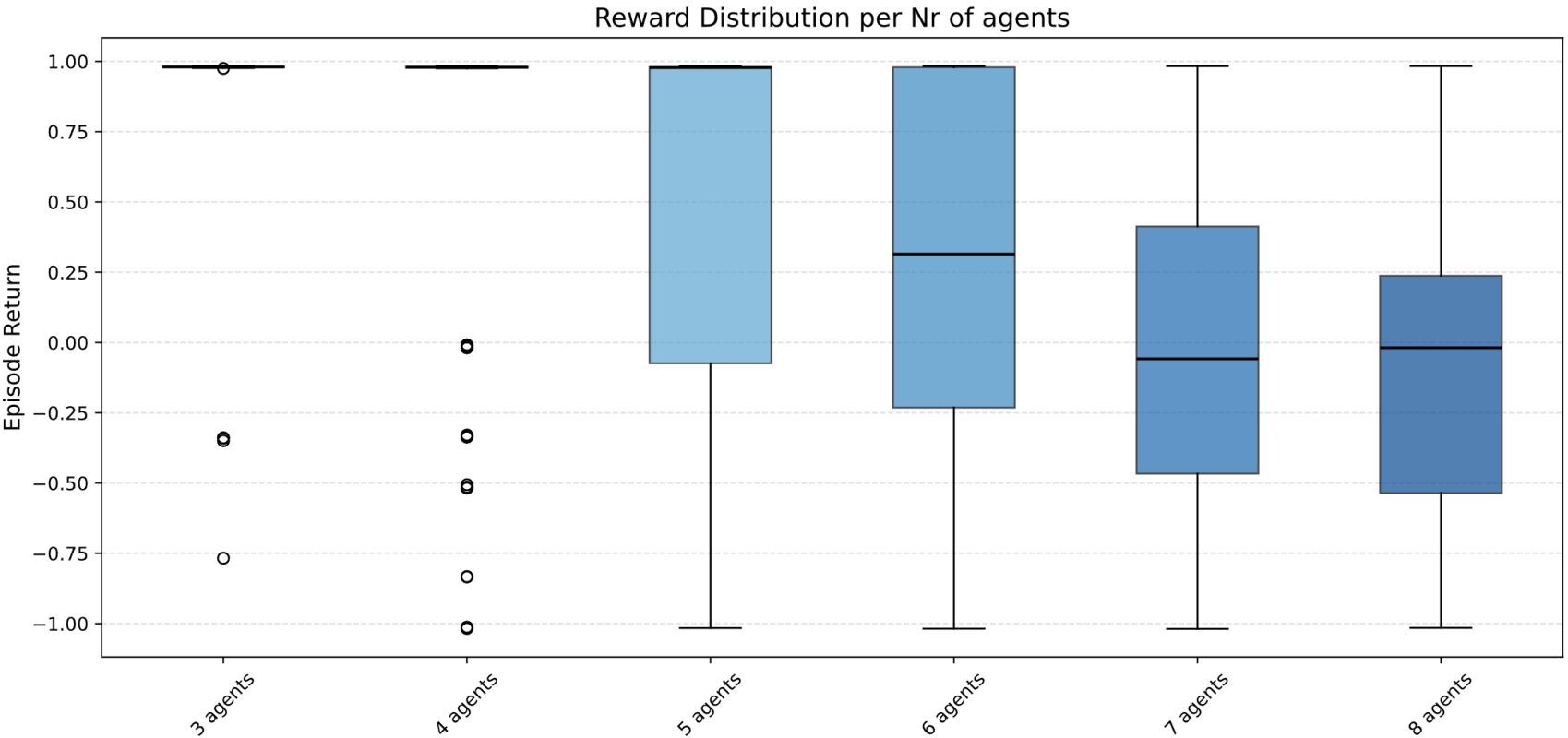
IPPO

Zero-Shot Generalization - Intersection



MAPPO

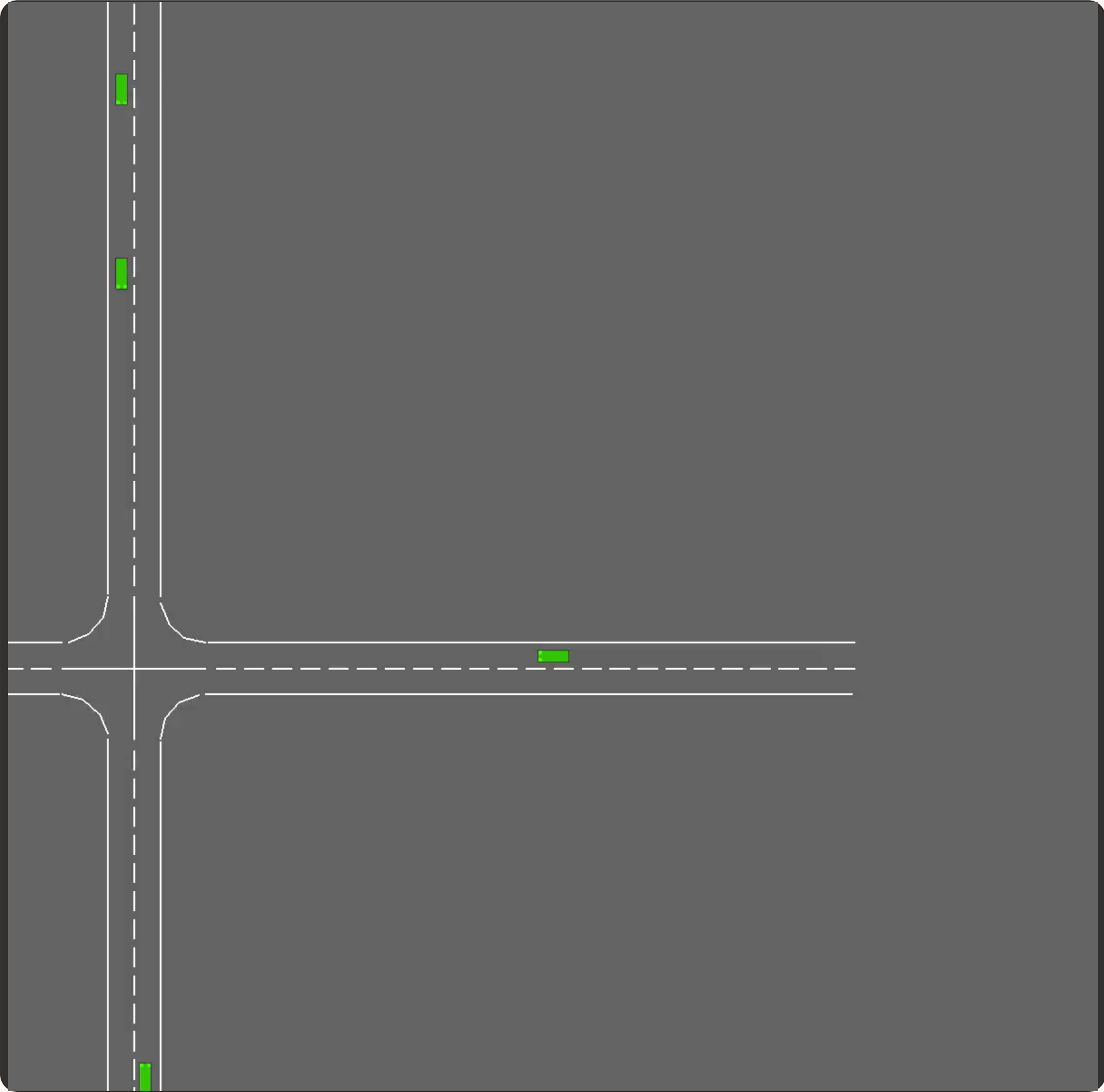
Zero-Shot Generalization - Intersection



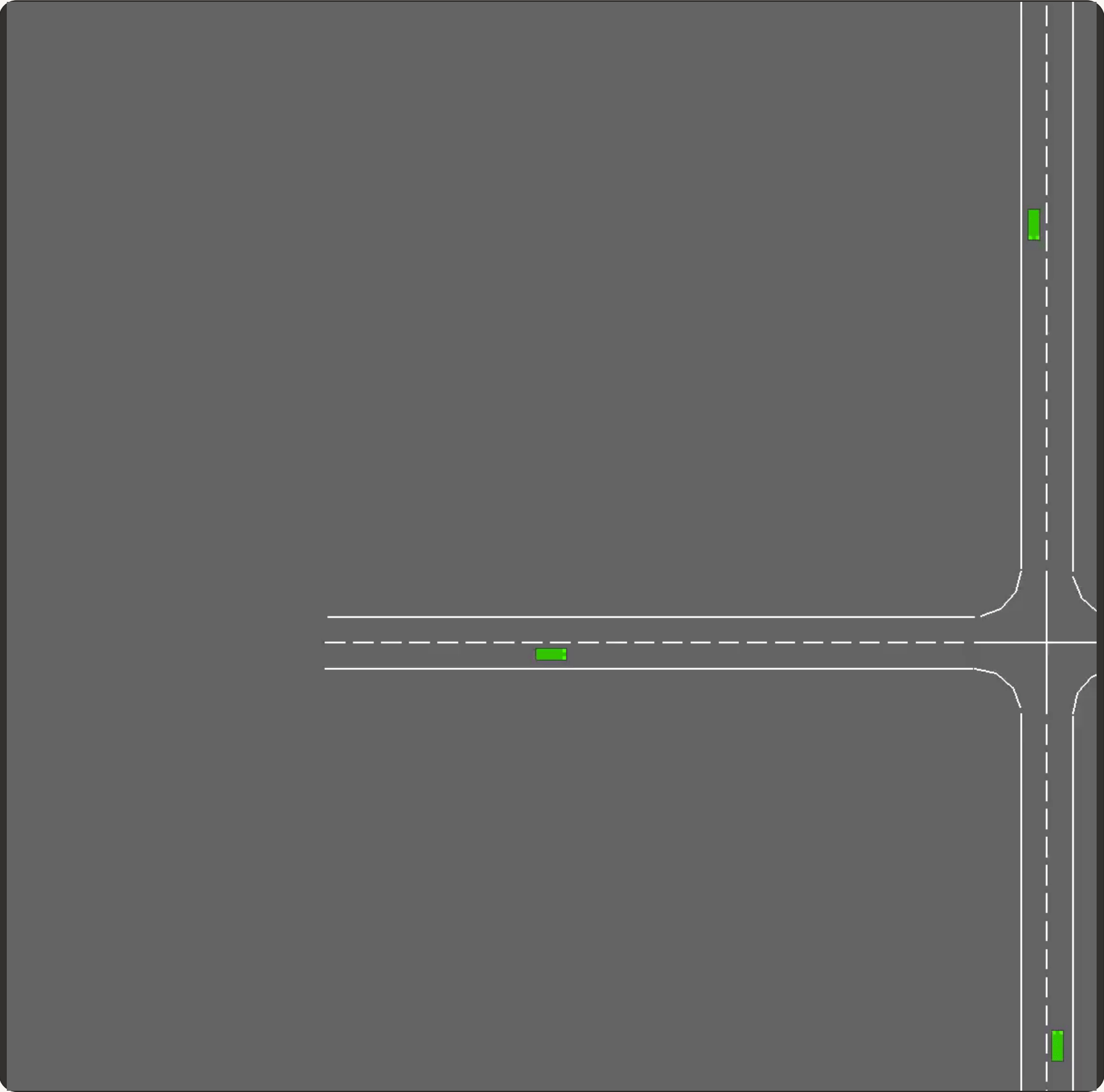
The centralized critic **brings no advantage** where reacting to the local state is enough. [1] de Witt et al. 2020 · [2] Yu et al. 2022, equivalence on part of SMAC.

Crossing behavior compared

IPPO



MAPPO



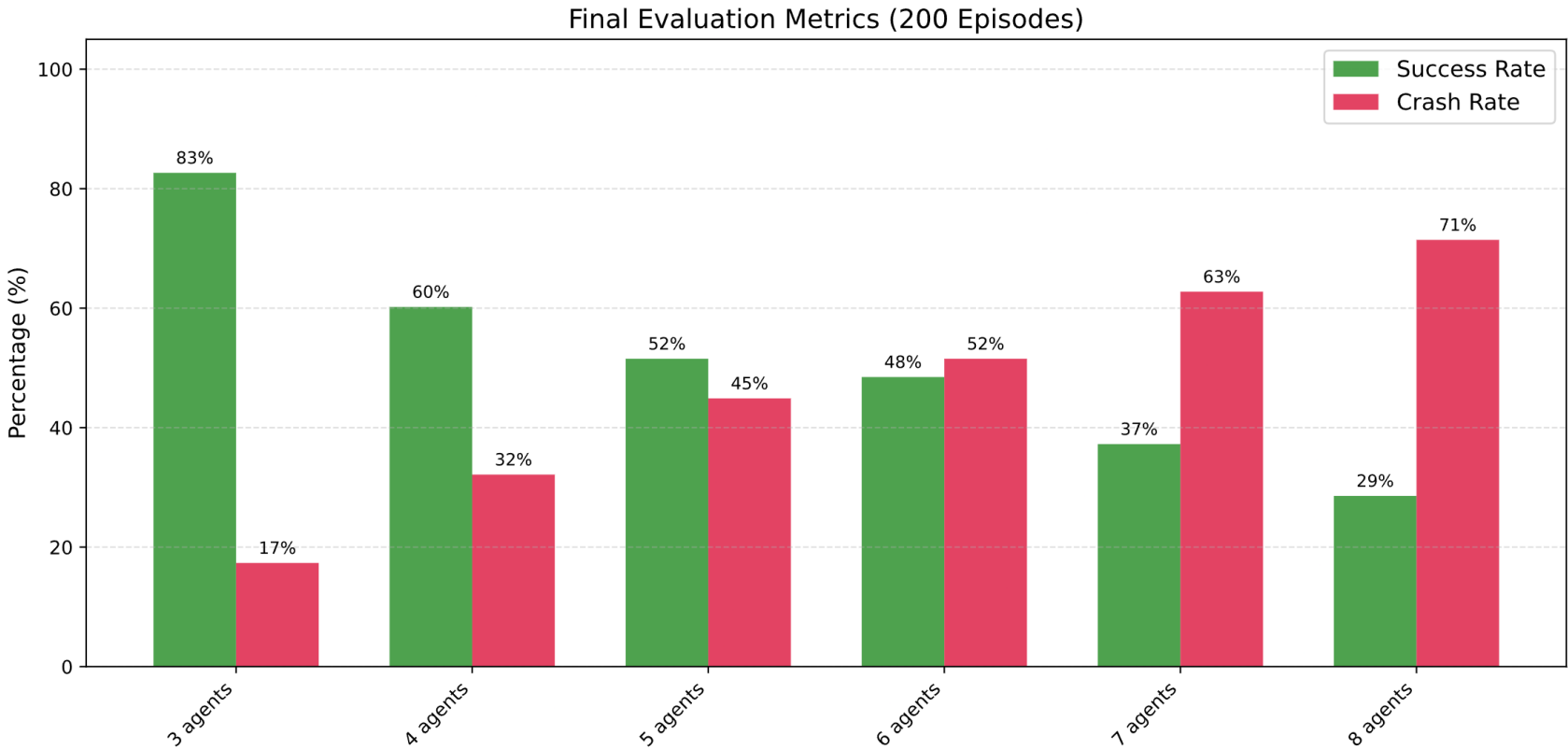
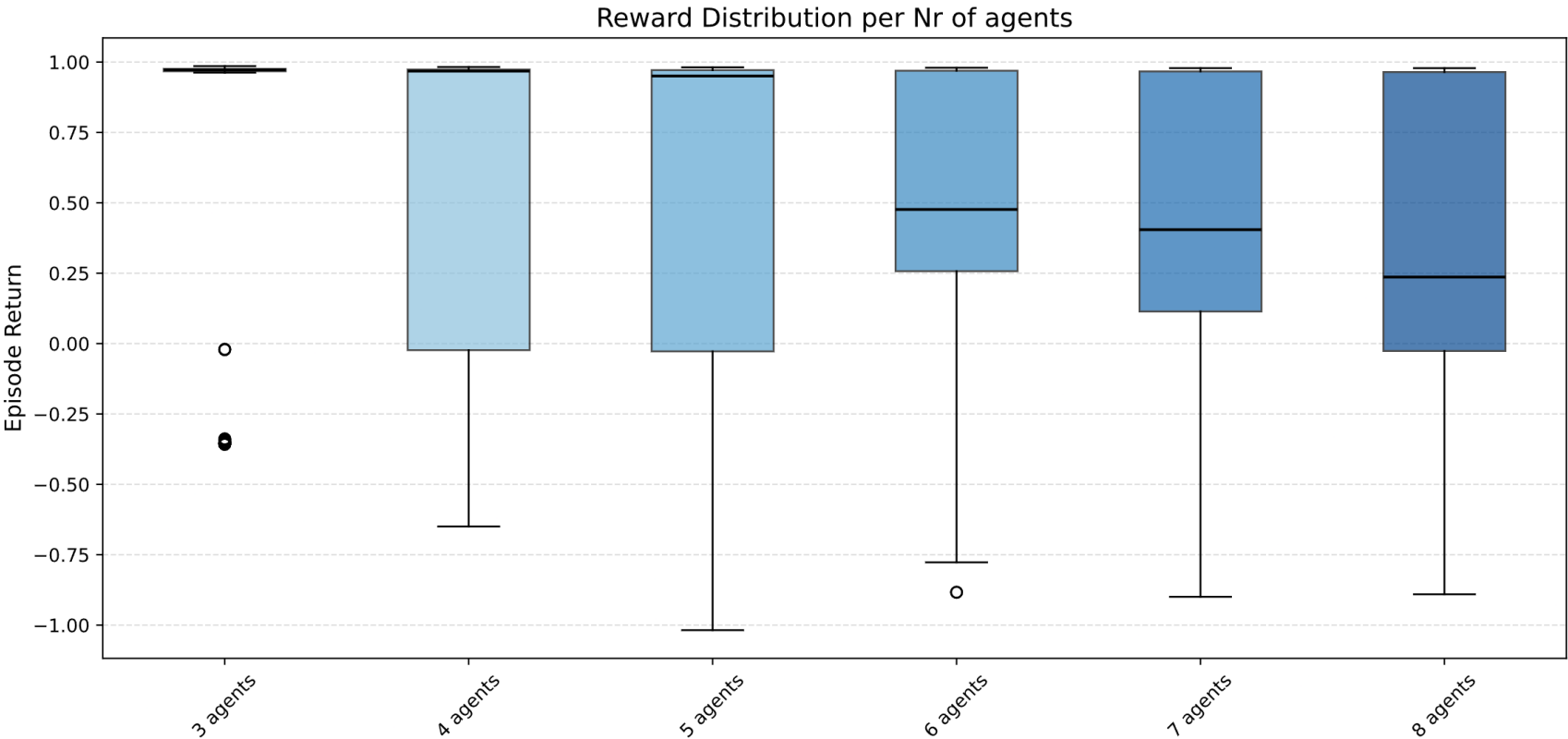
At low density both cross reliably. As the number of vehicles increases, **even the best policies end up colliding**: the intersection remains chaotic and reactive.

IPPO vs MAPPO

Evaluation on Roundabout

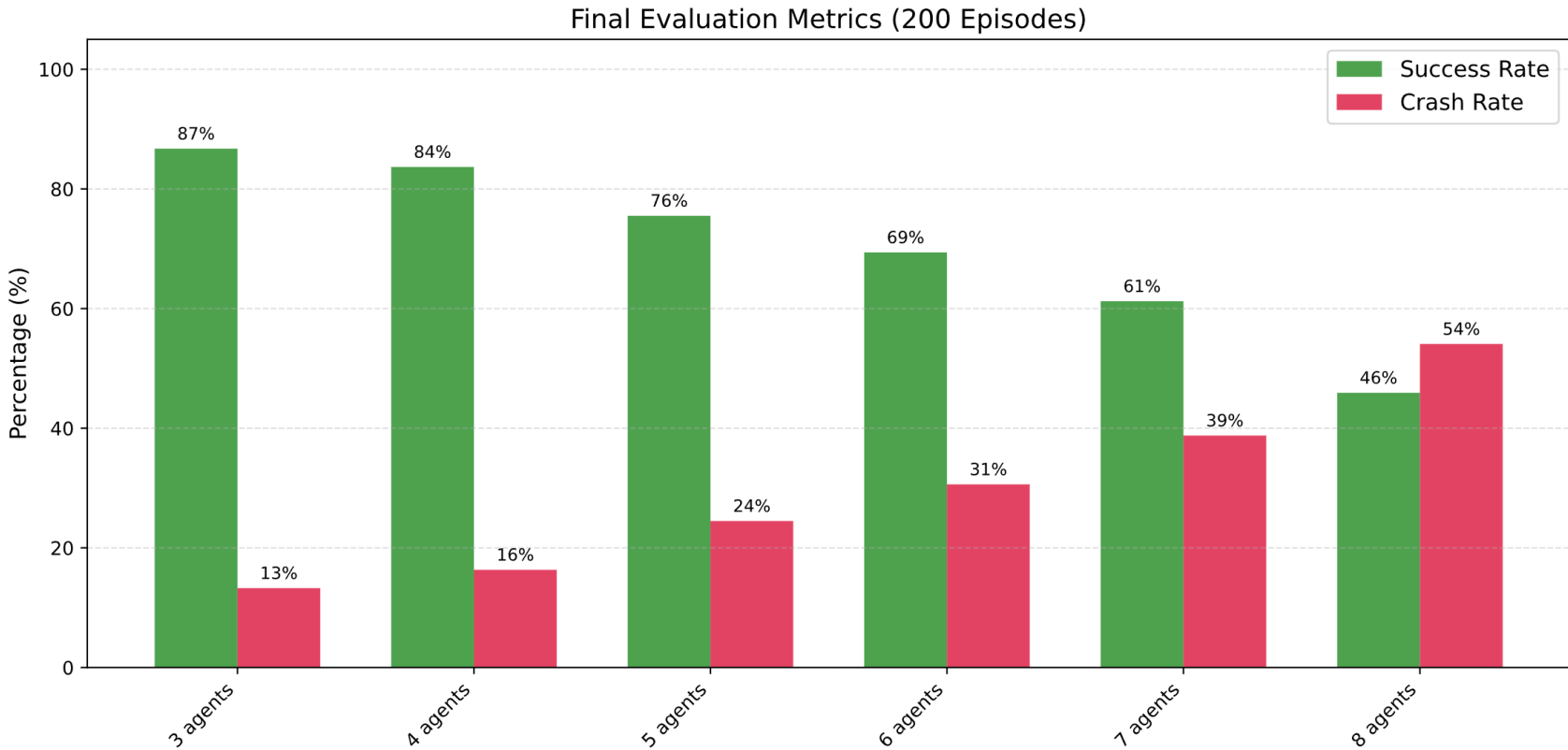
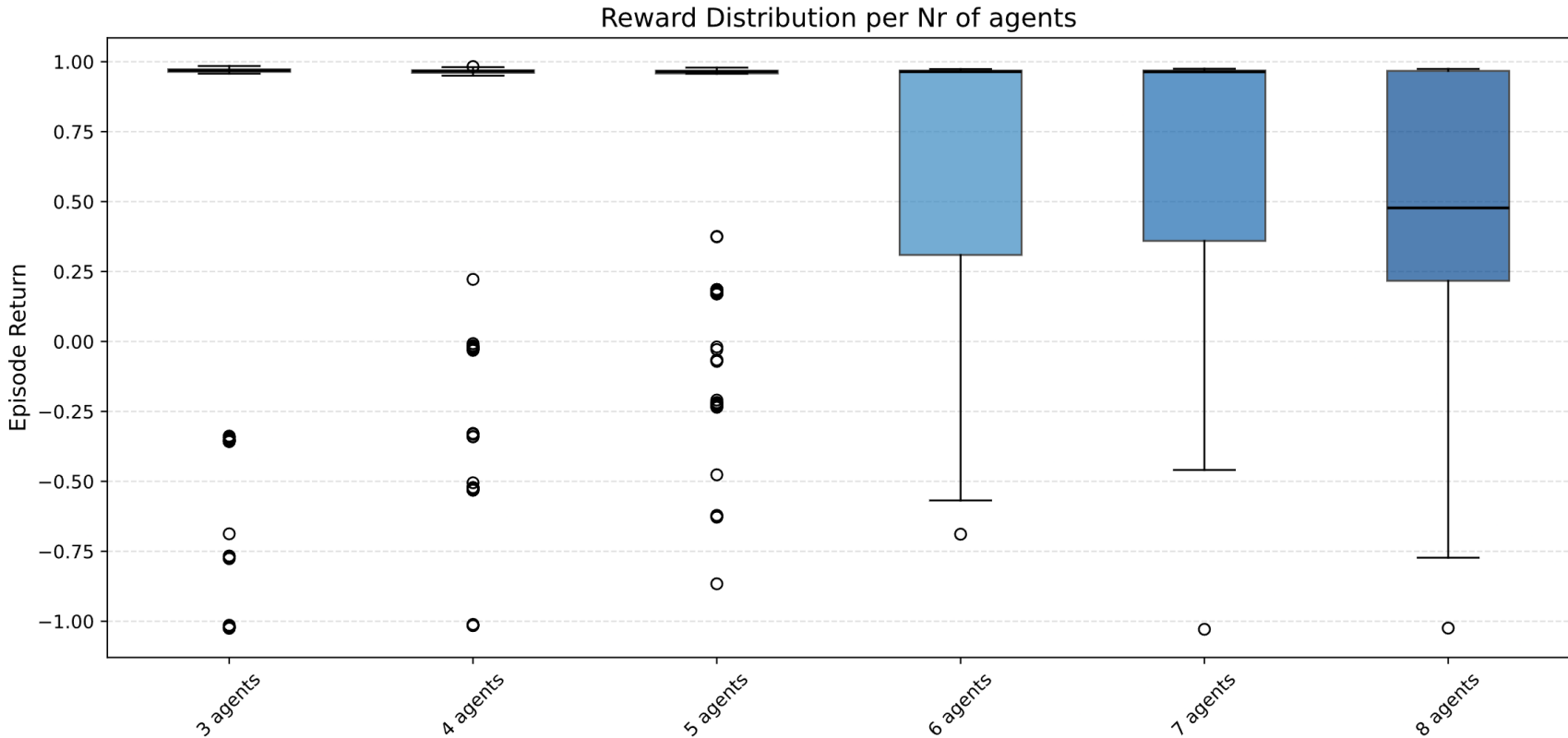
IPPO selfish → congestion

Zero-Shot Generalization - Roundabout - IPPO



MAPPO yields right of way

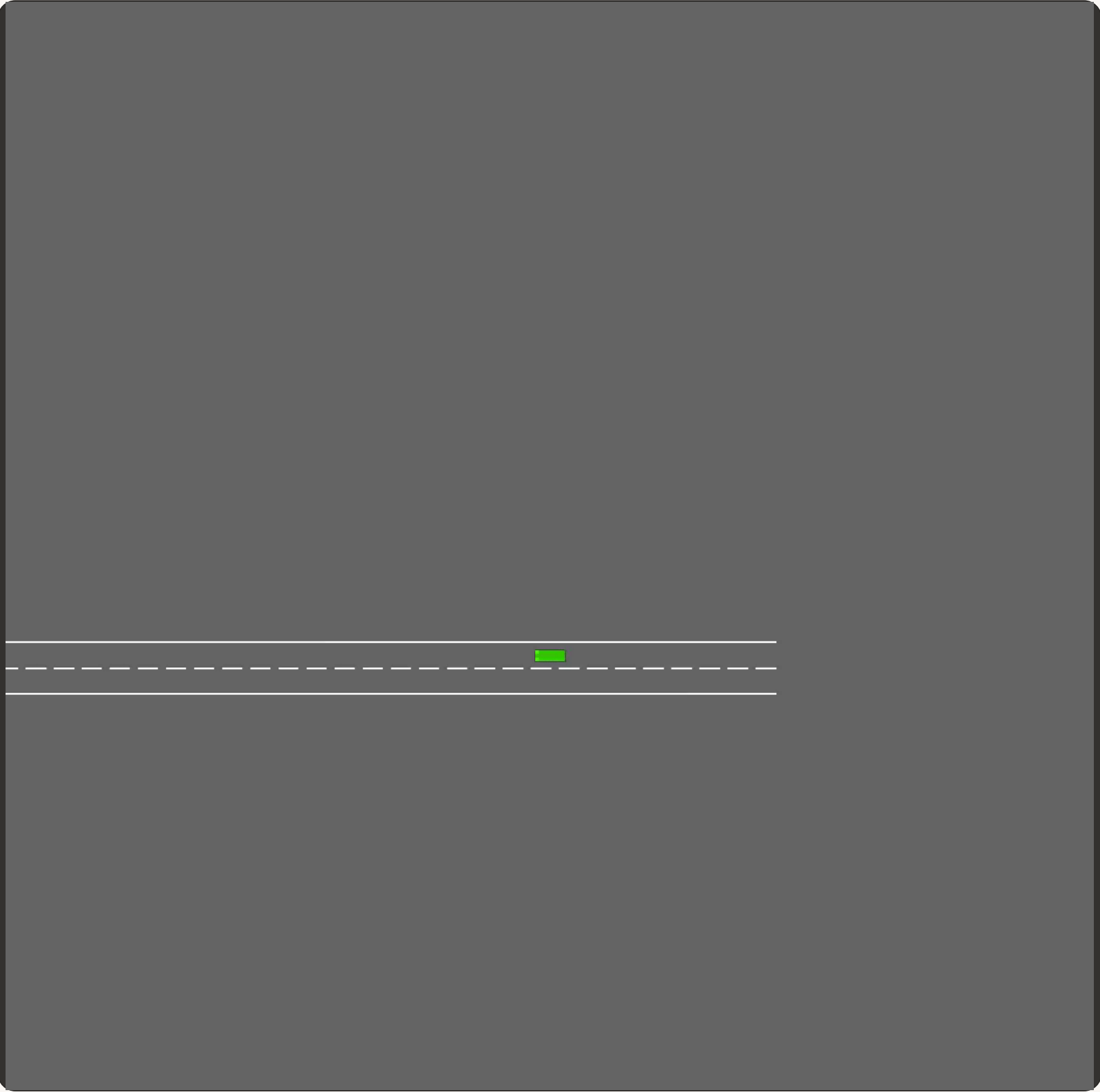
Zero-Shot Generalization - Roundabout - MAPPO



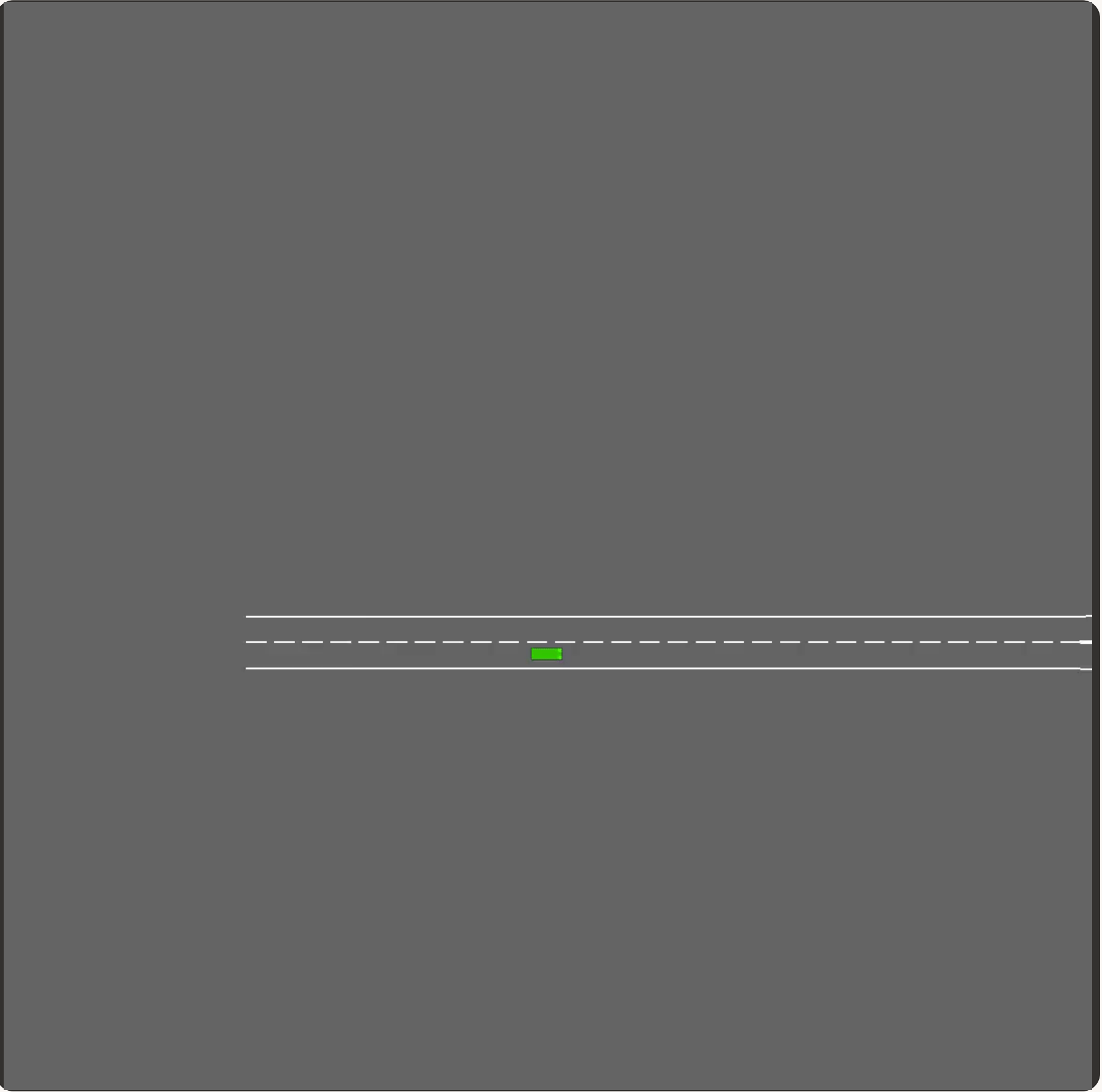
MAPPO dominates at every density; IPPO competes for the gap and collides. Cooperation emerges from the critic, not from the reward.

IPPO collides, MAPPO yields right of way

IPPO selfish → collision



MAPPO yields right of way → success



The roundabout requires **yielding**: MAPPO has internalized this behavior from the centralized critic, IPPO insists on the gap and collides.

MAPPO vs IPPO comparison

	Intersection chaotic · reactive	Roundabout structured · social
IPPO DTDE	Equivalent to MAPPO, with slightly more robustness at high density.	Selfish policy → causes congestion.
MAPPO CTDE	The Critic Centralized brings no advantage	Transfers cooperation.

the advantage of the critic centralized is **problem-dependent**.

Conclusions

1 The off-policy method tested **is not suitable** for our discrete, chaotic environment. Provisional verdict, requires further verification.

2 On the intersection the centralized critic **brings no advantage** : the two on-policy methods are equivalent, IPPO is sufficient.

3 MAPPO is **essential for transferable social behavior** : the critic teaches a cooperative semantics that generalizes to the roundabout.

4 The advantage of the centralized critic emerges **only where cooperation matters** . The price of decentralization: noisier value estimation in early training.

Limitations in the results

01

The limit of the MLP

Fixed-size, order-sensitive input. Truncating to the k nearest neighbors makes the agent blind beyond k. All density results are constrained by this.

→ **a size-invariant and permutation-invariant model is the next step.**

02

Fixed training density

Training always uses 3 agents. A curriculum over the number of agents was explored, but it is not the basis of the final comparison.

03

MASAC not optimized

A fair verdict on the off-policy method would require a dedicated hyperparameter search. For this reason the judgment remains provisional.

Essential bibliography

[1] C. Schröder de Witt et al. -- Is Independent Learning All You Need in the StarCraft Multi-Agent Challenge?
arXiv:2011.09533 · 2020

[2] C. Yu et al. -- The Surprising Effectiveness of PPO in Cooperative Multi-Agent Games
NeurIPS 2022 · arXiv:2103.01955



Code and materials: GitHub repository

<https://github.com/EmGira/Highway-env-MARL>

Thank you for your attention.